Oncolytic Virus Therapy

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1 Background

Oncolytic Virus Therapy is a method of immunotherapy that fights cancer through the use of genetically engineered viruses [1, 2]. Current methods take a pre-existing virus, for instance the somewhat commonly-used T-VEC herpes virus [1], remove the genes that cause the virus to dangerously attack healthy cells, and replace these genes with certain genes that increase the likelihood of the virus attacking cancer cells or accelerate the process of the immune system defending the body. This treatment is not without side effects, however, as it is extremely possible for the oncolytic virus to incite more aggressive involvement of the immune system that damages healthy cells, and it is potentially even more likely that the virus itself attacks healthy cells [1, 8].

The idea of this treatment began to gain some traction in the early 20th century, with some attempts to force viruses to evolve in the direction necessary for this treatment to be effective in the middle 20th century [3]. However, it was not until 2015 that the U.S. Food and Drug Administration approved the first oncolytic virus immunotherapy treatment for cancer [1, 13]. Therefore, as much of the viral genome is still not understood, significant research is still being done to determine the safest and most effective genome to use in the treatment.

In this paper, we train an AI model to generate viral DNA sequences to be used in oncolytic viral therapy.

2 Methods

To generate the oncolytic viral genome, we used 1,000 annotated viral genomes from the National Center for Biotechnology Information's nucleotide database [5]. We developed a naive simulation that, based on the annotations of the genes in each virus' genome, predicted whether that virus would have a positive effect on a person when placed in a cancerous organ. The genetic annotations were split into several categories: immune evasion genes (such as ICP34.5), which signify how well the virus is able to evade the immune system [6]; replication genes (such as helicase genes), which accelerate reproduction; natural tumor tropism factors (such as Rb binding genes), which encourages viruses to target tumor cells; and cell entry genes (such as spike) [9], which facilitates the virus' ability to attack healthy cells. These genes are then used to naively compute the probabilities of certain interactions between the viruses and different types of cells such as cancer cells, healthy cells, or immune cells. A simple loss function is then computed to determine the effect of the virus on the cancerous environment.

$$\begin{cases} \text{loss} = \max(0, \max(0, \alpha - \sigma S) + \lambda V - \delta I) & \text{if cancer wins} \\ \text{loss} = \max(0, \max(0, \beta + \sigma S) + \lambda V - \delta I) & \text{if cancer loses} \end{cases}$$

Where the constants are:

- $\alpha :=$ baseline loss for cancer winning (we used 1000)

- $\beta :=$ baseline loss for cancer losing (we used -100)
- $\sigma :=$ coefficient for each timestep it takes for the winner to emerge (we used 10)
- $\lambda :=$ coefficient for the number of viruses (we used $\frac{1}{50}$)
- $\delta :=$ coefficient for the number of remaining immune cells (we used 1)

And the variables are:

- $S :=$ number of timesteps for the winner to emerge
- $V :=$ number of final final viruses
- $I :=$ number of remaining immune cells

Following the simulation of each viral genome, we fed the genome sequence and its loss into a LSTM (Long Short-term Memory) RNN (Recurrent Neural Network). The network had 3 layers with ReLU activation functions after the first two. The model used the Adam optimization algorithm to optimize for cross-entropy loss. It also kept track of the rate of appearance of different codons weighed by the loss of the entire sequence and used this to inherently learn relationships between combinations of codons and the virus' performance in the cancerous environment. Using this learned relationship the LSTM then generates a new DNA sequence and estimates the new sequence's loss based on its training data.

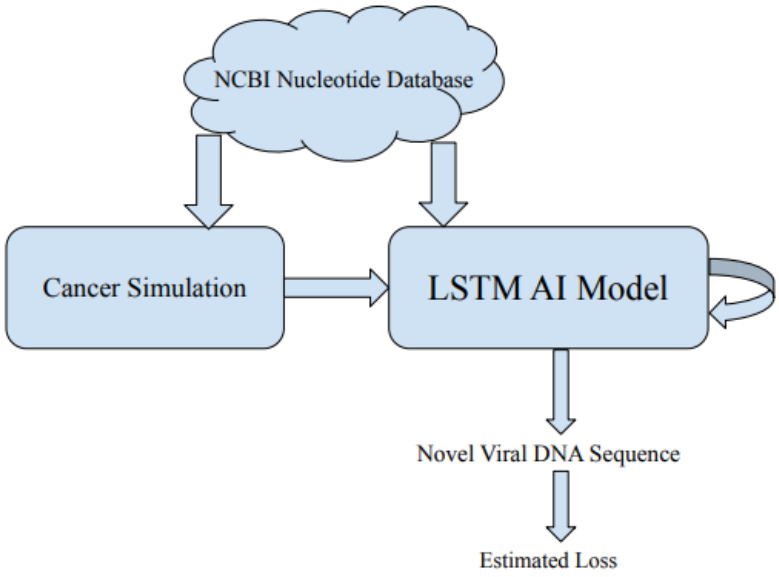


Figure 1: Full experiment pipeline

3 Results

Out of the $1,000$ viral genotypes that we ran through the simulation, the one that performed the best in the simulation was Alphainfluenzavirus with a loss of only 11.92 (although many of the most common viruses used in oncolytic viral therapy were not used). However, the novel generated sequence performed significantly worse than the Alphainfluenza with a loss of 5882.22. Despite this, the novel virus was still considerably better than most of the viruses 2, with many of the viruses having losses greater than $10,000$.

Therefore, it is likely that the immune system would not be worsened by the virus, but it is unclear whether it would be beneficial.

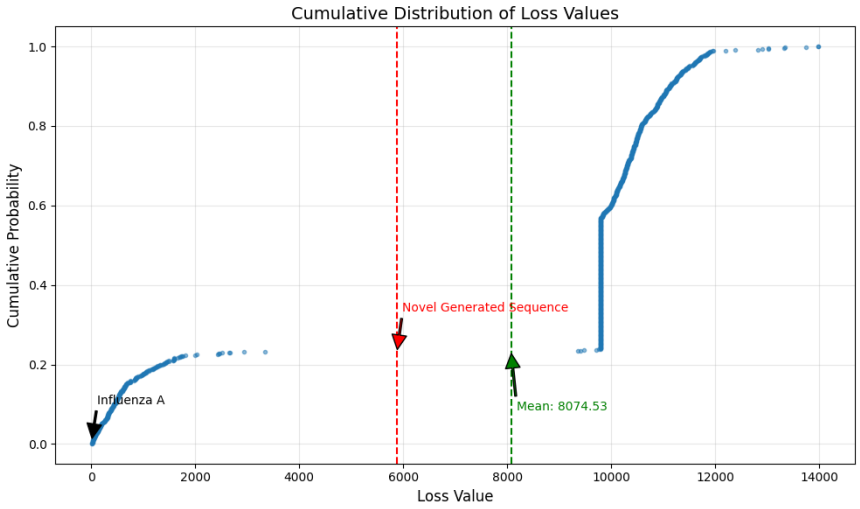


Figure 2: Cumulative distribution of losses of different viral genotypes. Two groups are clearly present: the group where the person beats cancer, and the group where the person does not. Our generated viral genotype leans towards beating cancer, while the mean leans towards losing.

The code for the experiment can be viewed on Github with the full generated sequence there as well.

4 Discussion

Much of this experiment was extremely naive and used as a way of gaining a mere understanding of the effects of different genes on the behavior of viruses and their interactions in the body. The simulation was very unrealistic and only took into account a few ways in which viruses interact with the human body. Additionally, the model used to generate the sequence was extremely weak, and likely had a difficult time learning much of substance, and in this type of experiment, an extremely deep neural network would likely be optimal. Additionally, the final generated DNA was not ran through the simulation because annotation of the genes would be necessary. Because there is no existing streamlined computational library for genome annotation, this is difficult and beyond the scope of this paper; however, it would be an interesting continuation of this experiment.

Despite this naivete, the experiment presented an interesting result; sections of the Influenza A virus have many of the genes that make viruses effective at fighting cancer, which is worth researching further. The simulation may not have been extremely accurate, but it is certainly competent at pinpointing key genes, and it did this with the Influenza A virus. Additionally, this pipeline has significant potential if built out a bit more with more realistic simulations, a more powerful model, and a fleshed out reinforcement learning pipeline to aid the model in generating more effective viral DNA. Finally, with a streamlined DNA annotation library, more thorough experiments than this have significant potential to uncover useful patterns in DNA that can be used in practice through oncolytic virus therapy and other interesting fields.

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